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CIRCULAR LINKED LIST

PROGRAM CODING:

```
#include<stdio.h>
#include<malloc.h>
struct node
{
  int info;
  struct node *next;
}*h;
void create(struct node *);
void insertfront(struct node *,int);
void insertmiddle(struct node *,int);
void insertend(struct node *,int);
void deletefront(struct node *);
void deletemiddle(struct node *);
void deleteend(struct node *);
void search(struct node *);
void display(struct node *);
void main()
{
  struct node *h;
  int n,s,op;
  clrscr();
  h=(struct node *)malloc(sizeof(struct node *));
```

```
 printf("\n\n\n\n\t\t\t\t Linked list operation");

 printf("\n\n\t1.Create\n\t2.Insert front\n\t3.Insert middle\n\t4.Insert end\n");

 printf("\n\t5.Delete front\n\t6.Delete middle\n\t7.Delete end\n\t8.search\n\t9.Display\n\t10.Exit\n");

 while(1)

 {

  printf("\n Enter your option:\t");

  scanf("%d",&op);

  switch(op)

  {

   case 1:create(h);break;

   case 2:

        printf("\n Enter the number to insert:\t");

        scanf("%d",&n);

        insertfront(h,n);

        break;

   case 3:

        printf("\n Enter the number to insert:\t");

        scanf("%d",&n);

        insertmiddle(h,n);

        break;

   case 4:

        printf("\n Enter the number to insert:\t");

        scanf("%d",&n);

        insertend(h,n);

        break;

   case 5:
```

```
            deletefront(h);
            break;
        case 6:
            deletemiddle(h);
            break;
        case 7:
            deleteend(h);
            break;
        case 8:
            search(h);
            break;
        case 9:
            display(h);
            break;
        case 10:
             exit(0);
             break;
        default:printf("\n.....Invalid option\n");
      }
   }
}
void create(struct node *h)
{
  int i,n,x;
  struct node *s1,*s2;
  s1=(struct node *)malloc(sizeof(struct node *));
  printf("\n Enter the number of elements:\t");
```

```
  scanf("%d",&x);
  printf("\n Enter the element:\t");
  scanf("%d",&n);
  s1->info=n;
  s1->next=h;
 h->next=s1;
  for(i=0;i<x-1;i++)
  {
    s2=(struct node *)malloc(sizeof(struct node *));
    s1->next=s2;
    printf("\n Enter the element:\t");
    scanf("%d",&n);
    s2->info=n;
    s2->next=h;
    s1=s1->next;
  }
  display(h);
  printf("\n");
}
void insertfront(struct node *h,int n)
{
  struct node *x;
  x=(struct node *)malloc(sizeof(struct node *));
  x->info=n;
  x->next=h->next;
  h->next=x;
  display(h);
```

```
  printf("\n");
}
void insertmiddle(struct node *h,int n)
{
  struct node *x,*y;
  int e;
  x=(struct node *)malloc(sizeof(struct node *));
  x->info=n;
  x->next=h;
  printf("\n Enter the element after which the given element to be inserted:\t");
  scanf("%d",&e);
  y=h;
  while(y->info!=e)
  y=y->next;
  x->next=y->next;
  y->next=x;
  display(h);
}
void insertend(struct node *h,int n)
{
  struct node *x,*y;
  x=(struct node *)malloc(sizeof(struct node *));
  x->info=n;
  x->next=h;
  y=h;
  while(y->next!=h)
  y=y->next;
```

```
 y->next=x;
 display(h);
}
void deletefront(struct node *h)
{
 if(h->next!=h)
 h->next=h->next->next;
 else
 h=h;
 display(h);
}
void deletemiddle(struct node *h)
{
 int e;
 struct node *x,*y;
 y=h;
 x=y->next;
 printf("\n Enter the element \t");
 scanf("%d",&e);
 while(x->info!=e)
  {
   x=x->next;
   y=y->next;
  }
 y->next=x->next;
 display(h);
}
```

```
void deleteend(struct node *h)
{
 struct node *x,*y;
 y=h;
 x=y->next;
 while(x->next=h)
  {
   x=x->next;
   y=y->next;
  }
 y->next=h;
 display(h);
}
void search(struct node *h)
{
 struct node *x,*y;
 int s;
 printf("\n Enter the number:\t");
 scanf("%d",&s);
 y=h;
 while(y->info!=s&&y->next!=h)
 y=y->next;
 if(y->info==s)
 printf("Element found\n");
 else
 printf("Element not found \n");
}
```

```
void display(struct node *h)
{
  struct node *y;
  y=h;
  printf("h->");
  while(y->next!=h)
  {
    y=y->next;
    printf("%d->",y->info);
  }
  printf("h");
}
```